

**German University in Cairo (GUC)**  
Media Engineering and Technology (MET)

# Automated Hallucination Detection Report

*Neuro-Symbolic Verification via Knowledge Graphs*

Bachelor Thesis Project Output

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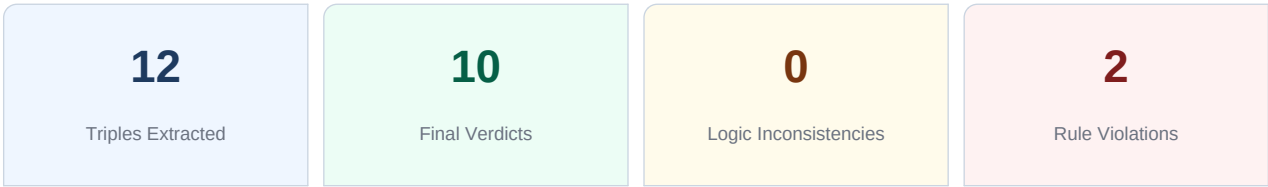
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REPORT OVERVIEW & METHODOLOGY

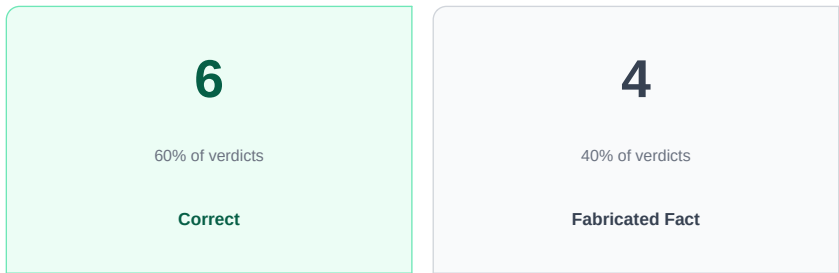
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

| #  | Subject | Predicate                        | Object                                                                | Type | Predicate Definition                                                                                  |
|----|---------|----------------------------------|-----------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------|
| 1  | 663     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 44%                                                                   |      | Specifies the highest allowable debt-to-income ratio for credit score 663.                            |
| 2  | 663     | HAS_LENDER_POOL                  | regional banks                                                        |      | Indicates the group of lenders available for credit score 663.                                        |
| 3  | 667     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 44%                                                                   |      | Specifies the highest allowable debt-to-income ratio for credit score 667.                            |
| 4  | 667     | HAS_LENDER_POOL                  | regional banks                                                        |      | Indicates the group of lenders available for credit score 667.                                        |
| 5  | 660     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 44%                                                                   |      | Specifies the highest allowable debt-to-income ratio for credit score 660.                            |
| 6  | 660     | HAS_LENDER_POOL                  | regional banks                                                        |      | Indicates the group of lenders available for credit score 660.                                        |
| 7  | 960     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | NOT_INCLUDED (Credit score 960 is outside the valid range of 300-850) |      | Specifies that maximum debt-to-income ratio does not apply for credit score 960 due to invalid score. |
| 8  | 960     | HAS_LENDER_POOL                  | NOT_INCLUDED (Credit score 960 is outside the valid range of 300-850) |      | Indicates that lender pool is not available for credit score 960 due to invalid score.                |
| 9  | 855     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 52%                                                                   |      | Specifies the highest allowable debt-to-income ratio for credit score 855.                            |
| 10 | 855     | HAS_LENDER_POOL                  | wealth management lenders                                             |      | Indicates the group of lenders available for credit score 855.                                        |
| 11 | 750     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 49%                                                                   |      | Specifies the highest allowable debt-to-income ratio for credit score 750.                            |
| 12 | 750     | HAS_LENDER_POOL                  | all major banks                                                       |      | Indicates the group of lenders available for credit score 750.                                        |

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

| Subject | Original Predicate               | Object         | Canonical Predicate                                                     | Conf.  |
|---------|----------------------------------|----------------|-------------------------------------------------------------------------|--------|
| 663     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 44%            | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | MEDIUM |
| 663     | HAS_LENDER_POOL                  | regional banks | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | MEDIUM |
| 667     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 44%            | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | MEDIUM |
| 667     | HAS_LENDER_POOL                  | regional banks | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | MEDIUM |
| 660     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 44%            | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | MEDIUM |
| 660     | HAS_LENDER_POOL                  | regional banks | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | MEDIUM |

| Subject | Original Predicate               | Object                                                                   | Canonical Predicate                                                     | Conf.  |
|---------|----------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|
| 960     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | NOT_INCLUDED<br>(Credit score 960 is outside the valid range of 300-850) | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | MEDIUM |
| 960     | HAS_LENDER_POOL                  | NOT_INCLUDED<br>(Credit score 960 is outside the valid range of 300-850) | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | MEDIUM |
| 855     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 52%                                                                      | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | MEDIUM |
| 855     | HAS_LENDER_POOL                  | wealth management lenders                                                | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | MEDIUM |
| 750     | HAS_MAXIMUM_DEBT_TO_INCOME_RATIO | 49%                                                                      | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | MEDIUM |
| 750     | HAS_LENDER_POOL                  | all major banks                                                          | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | MEDIUM |

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually\_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

| Label                  | Inconsistency Type | Subject | Predicate                                                              | Object                    | Constraint ID       | Explanation                                                                      |
|------------------------|--------------------|---------|------------------------------------------------------------------------|---------------------------|---------------------|----------------------------------------------------------------------------------|
| factually_inconsistent | RULE_VIOLATION     | 855     | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                       | 52%                       | c_max_dti_range     | Credit score 855 is above the maximum permitted (850), violating the constraint. |
| factually_inconsistent | RULE_VIOLATION     | 855     | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THISSCORE | wealth management lenders | c_lender_pool_range | Credit score 855 exceeds the valid range 300-850, violating the constraint.      |

PART B — Hallucination Detection Verdicts (Pass 2)

| Subject | Predicate                                                              | Object         | Verdict         | Label                  | Inconsistency Type | Match | Violated | Conf.            | Explanation                                                                                                |
|---------|------------------------------------------------------------------------|----------------|-----------------|------------------------|--------------------|-------|----------|------------------|------------------------------------------------------------------------------------------------------------|
| 663     | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THISSCORE | regional banks | Fabricated Fact | factually_inconsistent | FABRICATED_FACT    | NONE  | SUBJECT  | ●●●●●●●●<br>0.90 | Object+predicate exist in DB but no matching subject found; subject is fabricated.                         |
| 667     | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THISSCORE | regional banks | Fabricated Fact | factually_inconsistent | FABRICATED_FACT    | NONE  | SUBJECT  | ●●●●●●●●<br>0.90 | Object+predicate exist in DB but no matching subject found; subject is fabricated.                         |
| 663     | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                       | 44%            | Fabricated Fact | factually_inconsistent | FABRICATED_FACT    | NONE  | MULTIPLE | ●●●●●●●●<br>0.90 | A concrete value ('44%') is asserted but no DB evidence supports any value for this subject and predicate. |
| 667     | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                       | 44%            | Fabricated Fact | factually_inconsistent | FABRICATED_FACT    | NONE  | MULTIPLE | ●●●●●●●●<br>0.90 | A concrete value ('44%') is asserted but no DB evidence supports any value for this subject and predicate. |

| Subject | Predicate                                                               | Object                                                                | Verdict | Label                | Inconsistency Type | Match    | Violated | Conf.         | Explanation                                                                                                                                                                                      |
|---------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|---------|----------------------|--------------------|----------|----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 660     | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | 44%                                                                   | Correct | factually_consistent | —                  | STRICT   | —        | ●●●●●●●● 1.00 | The asserted value ('44%') matches semantically with the DB candidate value '44'.<br><br>Ev: subject: 660, predicate: MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING, object: 44               |
| 660     | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | regional banks                                                        | Correct | factually_consistent | —                  | SEMANTIC | —        | ●●●●●●●● 1.00 | The object ('regional banks') is a semantic match to the DB candidate value.<br><br>Ev: subject: 660, predicate: LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE, ob...  |
| 960     | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | NOT_INCLUDED (Credit score 960 is outside the valid range of 300-850) | Correct | factually_consistent | —                  | NONE     | —        | ●●●●●●●● 0.90 | The object signals absence for this subject and predicate combination.                                                                                                                           |
| 960     | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | NOT_INCLUDED (Credit score 960 is outside the valid range of 300-850) | Correct | factually_consistent | —                  | NONE     | —        | ●●●●●●●● 0.90 | The object signals absence for this subject and predicate combination.                                                                                                                           |
| 750     | MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING                        | 49%                                                                   | Correct | factually_consistent | —                  | STRICT   | —        | ●●●●●●●● 1.00 | The asserted value ('49%') matches semantically with the DB candidate value '49'.<br><br>Ev: subject: 750, predicate: MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING, object: 49               |
| 750     | LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE | all major banks                                                       | Correct | factually_consistent | —                  | SEMANTIC | —        | ●●●●●●●● 1.00 | The object ('all major banks') is a semantic match to the DB candidate value.<br><br>Ev: subject: 750, predicate: LENDER_POOL_ACCESSIBLE_AS_AVAILABLE_INSTITUTION_CATEGORY_FOR_THIS_SCORE, ob... |



VERDICT LEGEND

| Verdict            | Description                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Correct            | All three triple components verified against the knowledge graph.                                               |
| Rule Violation     | One or more components contradict database evidence rule.                                                       |
| Unverifiable       | No usable DB evidence found for any component of this triple.                                                   |
| Predicate Mismatch | Subject entity exists, but the stated predicate is not a valid relation.                                        |
| Fabricated Fact    | Triple asserts a relationship that is in db but the subject, object or subject+object does not supported by db. |